

Goal: Body temperature sensing

Sensor Choice: MAX30208 (Maxim Integrated, USA) digital temperature sensor

- **Design Overview** The concept is to have a new part stretch out from the right side of the original circuit, at the end there is a temperature sensor. The corresponding area of retainer wall can be made thinner or replace by other material to ensure better thermal efficiency.

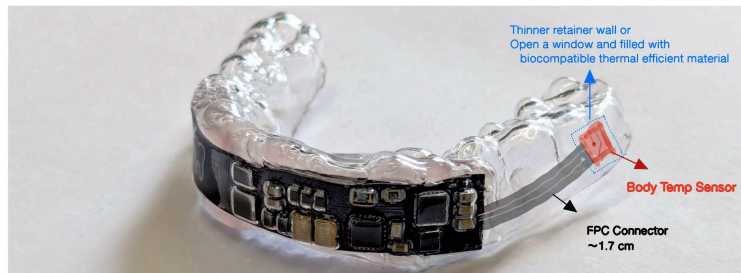


Figure 1: Concept Diagram (drawn using Notability)

- **Decision and Reasoning**

- **Placement Location:** For intra-oral body temperature measurement, the sensor should be placed in the posterior part of the oral cavity and farther from the mouth to reduce the influence of breathing, drinking... [1]. Combining with the current design of DENSOR, the sensor is chosen to be placed at the buccal side of the mandibular first molar, approximately 4 cm from the oral centerline. Assuming that DENSOR is placed at the centerline, a connector structure around 1.6 cm is included.
 - **Sensor Choice:** With reference also to Bueno et al.[1], MAX30208 (Maxim Integrated, USA) is chosen, as it is small, I2C-compatible, and has a clean and detailed datasheet...
 - **Circuit Details:** A 100 nF decoupling capacitor is placed next to the sensor, together forming a sensor island. A 1.6 cm connector structure leads to the sensor island, within which all four signal traces are constrained and guided by the ground plane. The design clearance was locally reduced to 0.075 mm to improve routing, requires confirmation against fabrication capabilities. Depending on fabrication and physical characteristics, the connector part can use a new board material. See the [evaluation system](#) for typical connection.
 - **Firmware Details:** The extended firmware was successfully compiled using STM32CubeIDE 2.2.0. Bit 4 of `sensorEnable` is now used to indicate the body temperature sensor. The sensor operates similarly to the existing ones, it is powered when the RTC activates and is started by `enableSensors()`. The recorded body temperature value is stored at the end of each measurement record and takes two bytes. As the sampling-ready time of the MAX30208 can be up to 50 ms, the firmware counts and allows a maximum of 50 iterative inquiries, with a 1 ms delay between each.
- **How It Works** After updating the NFC and Java part, the two-byte recording can be read and easily interpreted into degrees Celsius. The sensor recording has $\pm 0.1^{\circ}\text{C}$ accuracy over a $+30^{\circ}\text{C}$ to $+50^{\circ}\text{C}$ range, and a series of recordings over the last designated period makes it possible to evaluate body temperature. Extremely high and low results should be excluded.

Notes: KiCad 10.0 is used; version conflicts could happen.

Firmware is extended, but on the Java side the data structure length and number are no longer match. This need only small changes, overall it is highly integrated with the original framework.

[1] L. de Almeida e Bueno et al., “Determining the performance of a temperature sensor embedded into a mouthguard,” *BDJ Open*, 2022.